# Egyptian Adventure Fairness Verification

## Verification Parameters

| Parameter Name | Description |
|---|---|
| serverSeed | The server seed is generated by our system as a random 64-character hex string. |
| clientSeed | The client seed belongs to the player and is used to ensure that the player has influence on the outcomes' randomness. |
| nonce | The serial number of the bet |
| round | Rounds of scrolling |
| times | The number of scrolling in an inning |
| reelColumn | Sequence of the scroll reel |
| dropNumber | Number of symbols dropped |
| hash | Random number generated by HMAC_SHA256 |
| index | The territory of the intercepted hash |
| events | The selected random number value range [0,events) |

## Arithmetic

String hash = HMAC_SHA256 (string, serverSeed)

"double random = (Long.parseLong( hash.substring(8 * index, 8 * index + 14 ), 16) >> 3) * Math.pow(2, -53 ) * events;
**How to use:** random(hash, index, events)"

## Verification Steps

Place a bet -> Enter the <Base Reel> -> Start the <Eliminate Symbol>,
If the <Base Reel> Triggers free -> Start the <Free Reel> -> Start the <Elminate Symbol>, when the number of free times is 0, return to the <Base Reel> to continue to eliminate symbol.

Triggered by 3 or more Free Fall symbols plus a <WILD> symbol in a way. 3 Free Fall symbols in a win way award 9 Free Spins and each additional Free Fall symbol in a win way awards an extra 3 Free Spins)

## Base Reel

Initial parameters: round=1 / dropNumber=1

**Select scroll reels**
Select WILD reels
times=1
hash=HMAC_SHA256 (clientSeed:nonce:round:times, serverSeed)
index=0
events=351600, this value is equal to the sum of <The Base Game Weight column> in <The Reel Table Rule>
Replace the <Reel Table Rule> according to the result of random(hash, 0, 351600)

Select BASE reels
times=1
hash=HMAC_SHA256 (clientSeed:nonce:round:times, serverSeed)
index=1
events=2
Select <The Reel Table Rule> according to the result of random(hash, 1, 2)
If random=0, select Base Reel Set 1 in <The Reel Table Rule>
If random=1, select Base Reel Set 2 in <The Reel Table Rule>

**Show symbol number**
times=2
hash=HMAC_SHA256 (clientSeed:nonce:round:times, serverSeed)
index=[0,5]
events= the sum of one of column in the Base Game table in <The Reel Table Symbol Number>
eg: the number of symbol in the first column of the scroll reel, random(hash, 0, 130)

**Show symbol seat**
times=3
hash=HMAC_SHA256 (clientSeed:nonce:round:times, serverSeed)
index=[0,5]
events=The number of symbols in the current column of your selected scroll reel
eg: when the reel is <Base Reel Set>, index=0, then random(hash, 0, 275)

## Free Reel

Initial parameters: round=round+1 / dropNumber=1

**Select scroll reels**
Select WILD reels
times=1
hash=HMAC_SHA256 (clientSeed:nonce:round:times, serverSeed)
index=0
events=251600, this value is equal to the sum of <The Free Game Weight column> in <The Reel Table Rule>
Replace the reel table according to the result of random(hash, 0, 251600)

**Showing Symbol Number**
times=2
hash=HMAC_SHA256 (clientSeed:nonce:round:times, serverSeed)
index=[0,5]
events= the sum of one of column in the Free Game table in <The Reel Table Symbol Number>
eg: the number of symbol in the first column of the scroll reel, random(hash, 0, 50)

**Showing Symbol Seat**
times=3
hash=HMAC_SHA256 (clientSeed:nonce:round:times, serverSeed)
index=[0,5]
events=the number of symbols in the current column of the selected scroll reel
eg: When the reel is <Free Reel Set>, index=0, then random(hash, 0, 324)

## Eliminate Symbol

"When the reels start to eliminate the symbols
dropNumber=dropNumber+1
The reels will start the first time to eliminate if any symbols meets the conditions. It will enter the elimination round again after the first round of elimination is completed (if it can be eliminated again)

**How To Calculate Payline**
The reel will auto connect pay lines according to the rules and eliminate the symbols

**How To Calculate Free Reel**
If trigger <Free Reel> in <Base Reel>, it will enter <Free Reel> immediately;
After the end of scrolling <Free Reel>, it will return to the current symbol to eliminate <Scatter>;
If you trigger <Free Reel> in <Free Reel>, will eliminate <Scatter> and increase the number of free times.

**Drop Symbols**
When the number of symbols in a column of reels eliminated is less than the symbols that should be displayed, the symbols will be dropped in sequence according to the rules

times=times+1
hash=HMAC_SHA256 (clientSeed:nonce:round:times:reelColumn, serverSeed)
index=[0,6]
reelColumn=[0,5]
events= calculate the weight ratio according to <The Drop Symbol Rule Table> of the dropped symbols
random(hash, index, events)
eg: In the first scrolling, when there is 1 symbol in the first column after elimination, the amount of displayed symbols is 3, and the index is the amount of symbols in the current column, when the first drop starts, random(hash, 1, event).
when the second drop starts, random(hash, 2, event)

**Earthquake**
When the current round has not eliminated the symbol and has not triggered an earthquake,
The probability of triggering an earthquake in <Base Reel> is 1/75
The probability of triggering an earthquake in <Free Reel> is 1/35

times=times+1
hash=HMAC_SHA256 (clientSeed:nonce:round:times, serverSeed)
index=0
events=75 or 35
eg: In the <Base Reel>, random(hash, 0, 75), if random=0, then it will trigger an earthquake

After all [PIC6]/[PIC7]/[PIC8]/[PIC9] are eliminated by earthquake,
the symbols will be dropped sequentially according to the steps of <Drop Symbols> (times will not increase by 1 in this situation).

## Base Game will use <Base Reel Set 1> and <Base Reel Set 2>, each with a probability of 50%

### Base Reel Set 1

| | R1 | R2 | R3 | R4 | R5 | R6 |
|---|---|---|---|---|---|---|
| [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] |

### Base Reel Set 2

| | R1 | R2 | R3 | R4 | R5 | R6 |
|---|---|---|---|---|---|---|
| [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] |

### Free Reel Set

| | R1 | R2 | R3 | R4 | R5 | R6 |
|---|---|---|---|---|---|---|
| [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] | [illegible] |

### Wild Reel

| | R1 |
|---|---|
| [illegible] | [illegible] |

## Table Includes Wild Roller Replacement Weight

| R1 | R2 | R3 | R4 | R5 | R6 | Base Game Weight | Free Game Weight | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | 300,000 | 80,000 | 0 Wild Reel |
| Wild Reel | | | | | | 8,000 | 8,000 | 1 Wild Reel |
| | Wild Reel | | | | | 8,000 | 8,000 | |
| | | Wild Reel | | | | 8,000 | 8,000 | |
| | | | Wild Reel | | | 8,000 | 8,000 | |
| | | | | Wild Reel | | 8,000 | 8,000 | |
| | | | | | Wild Reel | 8,000 | 8,000 | |
| Wild Reel | | | | | Wild Reel | 500 | 500 | 2 Wild Reels |
| | Wild Reel | | | | Wild Reel | 500 | 500 | |
| | | Wild Reel | | | Wild Reel | 500 | 500 | |
| Wild Reel | | | | Wild Reel | | 400 | 400 | |
| | Wild Reel | | | Wild Reel | | 400 | 400 | |
| | | Wild Reel | | Wild Reel | | 400 | 400 | |
| Wild Reel | | | Wild Reel | | | 300 | 300 | |
| | Wild Reel | | Wild Reel | | | 300 | 300 | |
| | | Wild Reel | Wild Reel | | | 300 | 300 | |

Base Game will use <Base Reel Set 1> and <Base Reel Set 2>, each with a probability of 50%
After deciding to use <Base Reel Set N>, determine whether there is Wild Reel according to the Base Game Weight in the table above
If there is <Wild>, the selected 1 or 2 reels will use <Wild Reel> instead

eg: This spin draws to use <Base Reel Set 2>, and draws 2 rounds for the <Wild Reel> with a chance of <Wild>

| | | | | | |
|---|---|---|---|---|---|
| | | Wild Reel | | Wild Reel | |

Then the scrolling used by the 6 reels is like this:

| | | | | | |
|---|---|---|---|---|---|
| Base Reel Set2 -R1 | Base Reel Set2 -R2 | Wild Reel -R1 | Base Reel Set2 -R4 | Wild Reel -R1 | Base Reel Set2 -R6 |

Free Game will use <Free Reel Set>
Determine whether there is <Wild Reel> according to <The Free Game Weight> in the above table
If there is <Wild Reel>, the selected 1 or 2 reels will use <Wild Reel> instead

## Table Of Base Game Reel Amount Weight

| Symbol Amounts | R1 | R2 | R3 | R4 | R5 | R6 |
|---|---|---|---|---|---|---|
| 7 | 5 | 5 | 5 | 5 | 5 | 5 |
| 6 | 10 | 10 | 10 | 10 | 10 | 10 |
| 5 | 15 | 15 | 15 | 15 | 15 | 15 |
| 4 | 30 | 30 | 30 | 30 | 30 | 30 |
| 3 | 40 | 40 | 40 | 40 | 40 | 40 |
| 2 | 30 | 30 | 30 | 30 | 30 | 30 |

## Table Of Free Game Reel Amount Weight

| Symbol Amounts | R1 | R2 | R3 | R4 | R5 | R6 |
|---|---|---|---|---|---|---|
| 7 | 5 | 5 | 5 | 5 | 5 | 5 |
| 6 | 5 | 5 | 5 | 10 | 10 | 10 |
| 5 | 10 | 10 | 10 | 10 | 10 | 10 |
| 4 | 10 | 10 | 10 | 10 | 10 | 10 |
| 3 | 10 | 10 | 10 | 10 | 10 | 10 |
| 2 | 10 | 10 | 10 | 10 | 10 | 10 |

The sequence of the Reels that the symbol drops is from left to right and from R1 to R6
The initial drop symbol is determined according to <The Reel Table>. When the initial symbol has connected paylines and earned score, the new drop symbol after elimination is determined according to the weight as per below tables:

## Base Game non-initial drop symbol <basic weight>

| Symbol | Normal Reel weight | WILD Reel weight |
|---|---|---|
| PIC1 | 100 | |
| PIC2 | 100 | |
| PIC3 | 125 | |
| PIC4 | 125 | |
| PIC5 | 125 | |
| PIC6 | 150 | |
| PIC7 | 150 | |
| PIC8 | 150 | |
| PIC9 | 150 | |
| SCAT1 | 60 | 0 |
| WILD | 0 | 60 |

If there is a{SCAT1} or a {WILD}, then {SCAT1} or {WILD} weight will become 0.
If there are a{SCAT1} or a {WILD} in the symbols currently dropped,
the weight of {SCAT1} or {WILD} will become 0, in the undropped part of this drop.

## Base Game non-initial drop symbol <adjusted weight>

The <adjusted weight> here is adjusted for the current patterns in the previous round,so it will be applied to R2 ~ R6, R1 is not applicable (because R1 does not have the previous round)

| DropNumber(include with initial drop) | Weight adjustment |
|---|---|
| 2 | +45 |
| 3 | +30 |
| 4 | +15 |
| 5 | -25 |
| 6 | -50 |
| 7+ | -90 |

<= This means the weight is adjusted, after the symbol is eliminated for the first time

<adjusted weight> will only change {PIC1 ~ PIC9} 's symbols drop weight, will not change {SCAT1 / WILD}'s drop weight.

## Free Game non-initial drop symbol <basic weight>

| Symbol | Normal Reel weight | WILD Reel weight |
|---|---|---|
| PIC1 | 100 | |
| PIC2 | 100 | |
| PIC3 | 125 | |
| PIC4 | 125 | |
| PIC5 | 125 | |
| PIC6 | 150 | |
| PIC7 | 150 | |
| PIC8 | 150 | |
| PIC9 | 150 | |
| SCAT1 | 60 | 0 |
| WILD | 0 | 60 |

If there is a{SCAT1} or a {WILD}, then {SCAT1} or {WILD} weight will become 0.
If there are a{SCAT1} or a {WILD} in the symbols currently dropped,
the weight of {SCAT1} or {WILD} will become 0, in the undropped part of this drop.

## Free Game non-initial drop symbol <basic weight>

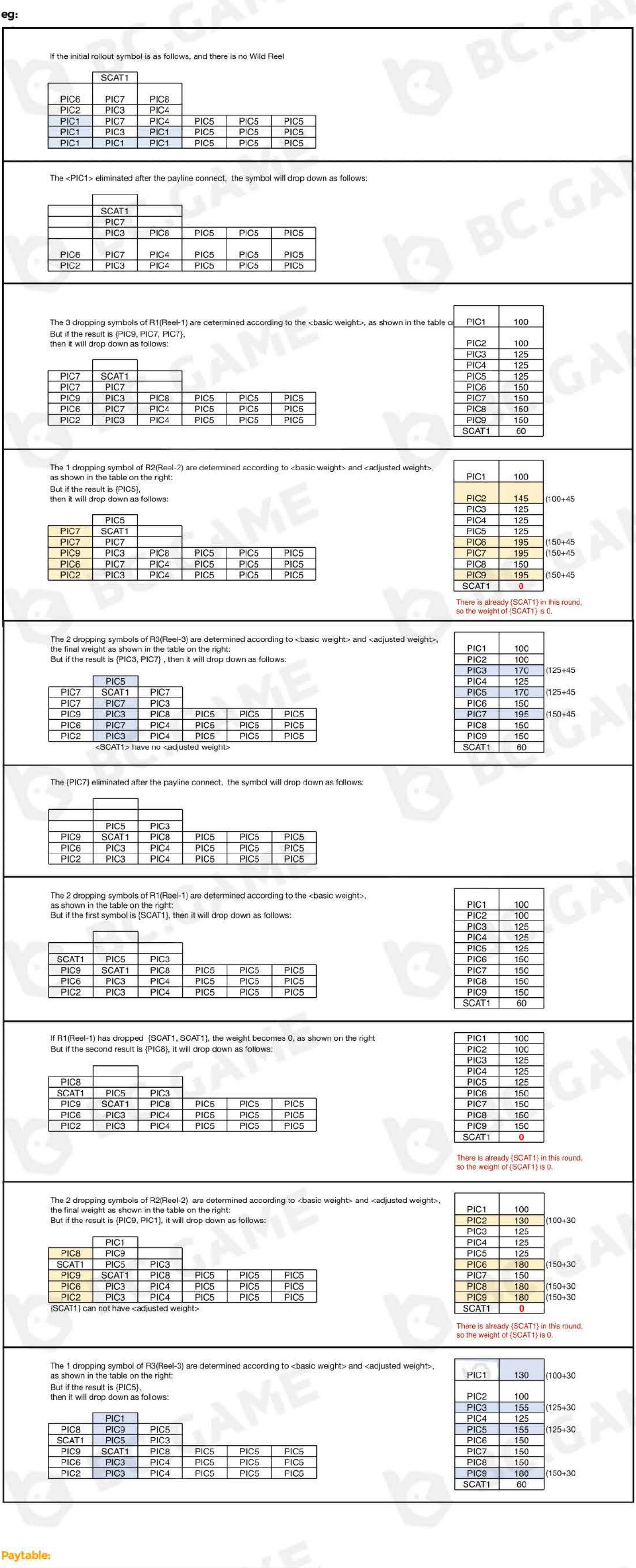
The <adjusted weight> here is adjusted for the current patterns in the previous round,
so it will be applied to R2 ~ R6, R1 is not applicable (because R1 does not have the previous round)

| DropNumber(include with initial drop) | Weight adjustment |
|---|---|
| 2 | +100 |
| 3 | +50 |
| 4 | +25 |
| 5 | 0 |
| 6 | -50 |
| 7+ | -90 |

<= This means the weight is adjusted, after the symbol is eliminated for the first time

<adjusted weight> will only change {PIC1 ~ PIC9} 's symbols drop weight, will not change {SCAT1 / WILD}'s drop weight.

**eg:**



If the initial rollout symbol is as follows, and there is no Wild Reel

| | | | | | |
|---|---|---|---|---|---|
| | SCAT1 | | | | |
| PIC6 | PIC7 | PIC8 | | | |
| PIC2 | PIC3 | PIC4 | | | |
| PIC1 | PIC7 | PIC4 | PIC5 | PIC5 | PIC5 |
| PIC1 | PIC3 | PIC1 | PIC5 | PIC5 | PIC5 |
| PIC1 | PIC1 | PIC1 | PIC5 | PIC5 | PIC5 |

The <PIC1> eliminated after the payline connect, the symbol will drop down as follows:

| | | | | | |
|---|---|---|---|---|---|
| | SCAT1 | | | | |
| | PIC7 | | | | |
| | PIC3 | PIC8 | PIC5 | PIC5 | PIC5 |
| PIC6 | PIC7 | PIC4 | PIC5 | PIC5 | PIC5 |
| PIC2 | PIC3 | PIC4 | PIC5 | PIC5 | PIC5 |

The 3 dropping symbols of R1(Reel-1) are determined according to the <basic weight>, as shown in the table o
But if the result is {PIC9, PIC7, PIC7},
then it will drop down as follows:

| | | | | | |
|---|---|---|---|---|---|
| PIC7 | SCAT1 | | | | |
| PIC7 | PIC7 | | | | |
| PIC9 | PIC3 | PIC8 | PIC5 | PIC5 | PIC5 |
| PIC6 | PIC7 | PIC4 | PIC5 | PIC5 | PIC5 |
| PIC2 | PIC3 | PIC4 | PIC5 | PIC5 | PIC5 |

| | |
|---|---|
| PIC1 | 100 |
| PIC2 | 100 |
| PIC3 | 125 |
| PIC4 | 125 |
| PIC5 | 125 |
| PIC6 | 150 |
| PIC7 | 150 |
| PIC8 | 150 |
| PIC9 | 150 |
| SCAT1 | 60 |

The 1 dropping symbol of R2(Reel-2) are determined according to <basic weight> and <adjusted weight>,
as shown in the table on the right:
But if the result is {PIC5},
then it will drop down as follows:

| | | | | | |
|---|---|---|---|---|---|
| | PIC5 | | | | |
| PIC7 | SCAT1 | | | | |
| PIC7 | PIC7 | | | | |
| PIC9 | PIC3 | PIC8 | PIC5 | PIC5 | PIC5 |
| PIC6 | PIC7 | PIC4 | PIC5 | PIC5 | PIC5 |
| PIC2 | PIC3 | PIC4 | PIC5 | PIC5 | PIC5 |

| | | |
|---|---|---|
| PIC1 | 100 | |
| PIC2 | 145 | (100+45 |
| PIC3 | 125 | |
| PIC4 | 125 | |
| PIC5 | 125 | |
| PIC6 | 195 | (150+45 |
| PIC7 | 195 | (150+45 |
| PIC8 | 150 | |
| PIC9 | 195 | (150+45 |
| SCAT1 | 0 | |

There is already {SCAT1} in this round,
so the weight of {SCAT1} is 0.

The 2 dropping symbols of R3(Reel-3) are determined according to <basic weight> and <adjusted weight>,
the final weight as shown in the table on the right:
But if the result is {PIC3, PIC7} , then it will drop down as follows:

| | | | | | |
|---|---|---|---|---|---|
| | PIC5 | | | | |
| PIC7 | SCAT1 | PIC7 | | | |
| PIC7 | PIC7 | PIC3 | | | |
| PIC9 | PIC3 | PIC8 | PIC5 | PIC5 | PIC5 |
| PIC6 | PIC7 | PIC4 | PIC5 | PIC5 | PIC5 |
| PIC2 | PIC3 | PIC4 | PIC5 | PIC5 | PIC5 |

<SCAT1> have no <adjusted weight>

| | | |
|---|---|---|
| PIC1 | 100 | |
| PIC2 | 100 | |
| PIC3 | 170 | (125+45 |
| PIC4 | 125 | |
| PIC5 | 170 | (125+45 |
| PIC6 | 150 | |
| PIC7 | 195 | (150+45 |
| PIC8 | 150 | |
| PIC9 | 150 | |
| SCAT1 | 60 | |

The {PIC7} eliminated after the payline connect, the symbol will drop down as follows:

| | | | | | |
|---|---|---|---|---|---|
| | | | | | |
| | PIC5 | PIC3 | | | |
| PIC9 | SCAT1 | PIC8 | PIC5 | PIC5 | PIC5 |
| PIC6 | PIC3 | PIC4 | PIC5 | PIC5 | PIC5 |
| PIC2 | PIC3 | PIC4 | PIC5 | PIC5 | PIC5 |

The 2 dropping symbols of R1(Reel-1) are determined according to the <basic weight>,
as shown in the table on the right:
But if the first symbol is {SCAT1}, then it will drop down as follows:

| | | | | | |
|---|---|---|---|---|---|
| SCAT1 | PIC5 | PIC3 | | | |
| PIC9 | SCAT1 | PIC8 | PIC5 | PIC5 | PIC5 |
| PIC6 | PIC3 | PIC4 | PIC5 | PIC5 | PIC5 |
| PIC2 | PIC3 | PIC4 | PIC5 | PIC5 | PIC5 |

| | |
|---|---|
| PIC1 | 100 |
| PIC2 | 100 |
| PIC3 | 125 |
| PIC4 | 125 |
| PIC5 | 125 |
| PIC6 | 150 |
| PIC7 | 150 |
| PIC8 | 150 |
| PIC9 | 150 |
| SCAT1 | 60 |

If R1(Reel-1) has dropped {SCAT1, SCAT1}, the weight becomes 0, as shown on the right
But if the second result is {PIC8}, it will drop down as follows:

| | | | | | |
|---|---|---|---|---|---|
| PIC8 | | | | | |
| SCAT1 | PIC5 | PIC3 | | | |
| PIC9 | SCAT1 | PIC8 | PIC5 | PIC5 | PIC5 |
| PIC6 | PIC3 | PIC4 | PIC5 | PIC5 | PIC5 |
| PIC2 | PIC3 | PIC4 | PIC5 | PIC5 | PIC5 |

| | |
|---|---|
| PIC1 | 100 |
| PIC2 | 100 |
| PIC3 | 125 |
| PIC4 | 125 |
| PIC5 | 125 |
| PIC6 | 150 |
| PIC7 | 150 |
| PIC8 | 150 |
| PIC9 | 150 |
| SCAT1 | 0 |

There is already {SCAT1} in this round,
so the weight of {SCAT1} is 0.

The 2 dropping symbols of R2(Reel-2) are determined according to <basic weight> and <adjusted weight>,
the final weight as shown in the table on the right:
But if the result is {PIC9, PIC1}, it will drop down as follows:

| | | | | | |
|---|---|---|---|---|---|
| | PIC1 | | | | |
| PIC8 | PIC9 | | | | |
| SCAT1 | PIC5 | PIC3 | | | |
| PIC9 | SCAT1 | PIC8 | PIC5 | PIC5 | PIC5 |
| PIC6 | PIC3 | PIC4 | PIC5 | PIC5 | PIC5 |
| PIC2 | PIC3 | PIC4 | PIC5 | PIC5 | PIC5 |

{SCAT1} can not have <adjusted weight>

| | | |
|---|---|---|
| PIC1 | 100 | |
| PIC2 | 130 | (100+30 |
| PIC3 | 125 | |
| PIC4 | 125 | |
| PIC5 | 125 | |
| PIC6 | 180 | (150+30 |
| PIC7 | 150 | |
| PIC8 | 180 | (150+30 |
| PIC9 | 180 | (150+30 |
| SCAT1 | 0 | |

There is already {SCAT1} in this round,
so the weight of {SCAT1} is 0.

The 1 dropping symbol of R3(Reel-3) are determined according to <basic weight> and <adjusted weight>,
as shown in the table on the right:
But if the result is {PIC5},
then it will drop down as follows:

| | | | | | |
|---|---|---|---|---|---|
| | PIC1 | | | | |
| PIC8 | PIC9 | PIC5 | | | |
| SCAT1 | PIC5 | PIC3 | | | |
| PIC9 | SCAT1 | PIC8 | PIC5 | PIC5 | PIC5 |
| PIC6 | PIC3 | PIC4 | PIC5 | PIC5 | PIC5 |
| PIC2 | PIC3 | PIC4 | PIC5 | PIC5 | PIC5 |

| | | |
|---|---|---|
| PIC1 | 130 | (100+30 |
| PIC2 | 100 | |
| PIC3 | 155 | (125+30 |
| PIC4 | 125 | |
| PIC5 | 155 | (125+30 |
| PIC6 | 150 | |
| PIC7 | 150 | |
| PIC8 | 150 | |
| PIC9 | 180 | (150+30 |
| SCAT1 | 60 | |

## Paytable:

| | 3 | 4 | 5 | 6 |
|---|---|---|---|---|
| PIC1 | 1 | 2 | 5 | 15 |
| PIC2 | 0.7 | 1.2 | 2 | 4 |
| PIC3 | 0.5 | 0.7 | 1.2 | 2 |
| PIC4 | 0.4 | 0.6 | 1 | 1.2 |
| PIC5 | 0.3 | 0.5 | 0.8 | 1 |
| PIC6 | 0.2 | 0.4 | 0.6 | 0.8 |
| PIC7 | 0.2 | 0.3 | 0.5 | 0.7 |
| PIC8 | 0.2 | 0.3 | 0.4 | 0.6 |
| PIC9 | 0.1 | 0.2 | 0.3 | 0.5 |

BC.GAME